The Existence of Domino Magic Squares and Rectangles

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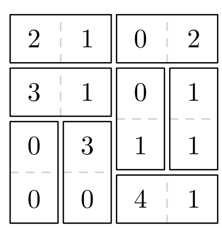
Abstract

An $r \times c$ domino magic rectangle is an arrangement of $rc/2$ dominoes to form an $r \times c$ rectangle such that every row has the same sum, and every column has the same sum. If $r = c$ and the two diagonals also have the same sum, it is a domino magic square. In this note we show that all possibilities for magic rectangles that are allowed by obvious arithmetical constraints actually exist. Further, we show that, considering rotations and reflections as distinct, there are 3639920 4×4 domino magic squares.

1 Introduction

A ***magic square*** is an arrangement of numbers such that every row, column and the two diagonals sum to the same value, called the ***magic sum***. A ***domino magic square*** is defined using a set of dominoes to form a magic square, each domino supplying two numbers. For example, here is a 4×4 domino magic square with 8 dominoes and magic sum 5. (It should be noted that there have also been examples in the literature where the two halves of the domino are added together to form one number: we do not consider this case.)



A ***domino magic rectangle*** is defined similarly, except that there are no diagonals, and only row and column sum constraints. Or rather, each row must add up to the same value, called the ***row-sum***, and each column must add up to the same value, called the ***column-sum***, but the row- and column-sums are different.

The concept of a domino magic square and rectangle has been looked at over the years. Most of these references have been in the form of puzzles. For example, Dudeney (see [2]) posed questions about 6×6 domino magic squares in his puzzle columns, while Duisenberg posed questions about the 4×4 domino magic square with minimum magic sum. They are also mentioned in puzzle books such as [4].

We could not find a known 7×8 magic rectangle, though Ball [1] has a nice arrangement where one outer row is all blank, and the remainder is a 7×7 magic square with magic sum 24. We assume throughout a standard double-6 set of 28 dominoes.

There are several obvious and well-known symmetries. For the sake of definiteness, we consider here rotations and reflections to be distinct squares/rectangles. There is also ***complementation***, where every pip is replaced by 6 minus the pip. So for example, if one has an $r \times c$ magic rectangle with row-sum R, one immediately gets an $r \times c$ magic rectangle with row-sum $6c - R$.

In this note we show that all possibilities for magic rectangles that are allowed by obvious arithmetical constraints actually exist. Further, we enumerate the 4×4 domino magic squares. Considering rotations and reflections as distinct, there are 3639920 domino magic squares. The data has some surprises.

2 Domino Magic Rectangles

We consider the existence of domino magic rectangles. Say the number of rows is r, the number of columns is c, the row-sum is R, and the column-sum is C. For definiteness, we assume $r \leq c$. Magic rectangles with only one row are rather boring, so we will assume $r \geq 2$.

2.1 Bounds from Arithmetic

What are the bounds? There are four natural constraints:

- Clearly rc is even.

- There are only 28 dominoes in the standard set, so one needs $rc \leq 56$. This means that the possible dimensions are: $r = 2$ and $2 \leq c \leq 28$; $r = 3$ and $3 \leq c \leq 18$; $r = 4$ and $4 \leq c \leq 14$; $r = 5$ and $5 \leq c \leq 11$; $r = 6$ and $6 \leq c \leq 9$; and $r = 7$ and $7 \leq c \leq 8$.

- The main bound on the row-sum is obtained by the sum of the pips. Define $f(i)$ to be the minimum pip-sum of i dominoes. For example, $f(4) = 5$, and is obtained with the 0-0, 0-1, 1-1 and 0-2 dominoes. It is necessary that the total sum of the pips is at least the pip-sum of $rc/2$ dominoes. That is,

$$r\,R \geq f(rc/2).$$

r c: possible R	2 20: 50,60	4 5: 10,15
2 3: 3,6,9	2 21: 63	4 6: 12,15,18
2 4: 4,6,8,10,12	2 22: 66	4 7: 14,21
2 5: 5,10,15	2 23: 69	4 8: 16,18,20,22,24
2 6: 6,9,12,15,18	2 24: 72	4 9: 27
2 7: 14,21	2 25: 75	4 10: 25,30
2 8: 12,16,20,24	2 26: 78	4 11: 33
2 9: 18,27	2 27: 81	4 12: 33,36
2 10: 15,20,25,30	2 28: 84	4 13: 39
2 11: 22,33	3 4: 4,8,12	4 14: 42
2 12: 24,30,36	3 6: 8,10,12,14,16,18	5 6: 12,18
2 13: 26,39	3 8: 16,24	5 8: 24
2 14: 28,35,42	3 10: 20,30	5 10: 28,30
2 15: 30,45	3 12: 28,32,36	6 6: 13,14,15,16,17,18
2 16: 40,48	3 14: 42	6 7: 21
2 17: 51	3 16: 48	6 8: 24
2 18: 45,54	3 18: 54	6 9: 27
2 19: 57	4 4: 5,6,7,8,9,10,11,12	7 8: 24

Table 1: The dimensions of domino magic rectangles with at most average row-sum

For example, $f(8) = 19$, and so for $r = c = 4$ one needs $R > 4$.

- Further, the row- and column-sums are related. So, for given R, one needs cR/r to be an integer.

It is easy to calculate the list of all possibilities. There are three values that the above conditions allow where the magic rectangle does not exist. The first is the 2×2 rectangle. It is easy to see that such an object would have every pip the same, which is impossible without duplicating a double domino.

The other two are the 2×16 domino magic rectangle with row-sum 32 and the 2×22 domino magic rectangle with row-sum 55. The first has a column-sum of 4 but requires dominoes with individual pips 5, which is a contradiction. The second has a column-sum of 5 but requires dominoes with individual pips 6, again a contradiction.

In Table 1 we list all possible domino magic rectangles where the average domino-pip is at most 3. That is, with $R \leq 3c$. For $R \geq 3c$, apply 6-complementarity to the rectangle with row-sum $6c - R$.

2.2 Domino Magic Rectangles with 2 Rows

Domino magic rectangles with two rows turn out to be simple. For, consider only the dimer tiling given by the dominoes. In order to fill the rectangle, the horizontal dominoes must come in pairs, one on top of the other. Further, in order to have such a magic rectangle with column-sum C, if there is a horizontal domino, the domino in the other row must be the C-complement. On the other hand, a vertical domino must be its own C-complement. Thus we really have a collection of vertical dominoes, and 2×2 squares, with just the constraint of getting the two row-sums to match.

Here for example is a 2×18 domino magic rectangle with row-sum 45 and column-sum 5.

0	2	4	4	0	0	4	4	5	0	3	3	4	0	1	5	2	4
5	3	1	1	5	5	1	1	0	5	2	2	1	5	4	0	3	1

2.3 Generating Other Domino Magic Rectangles

The magic rectangles with more rows were generated using an assortment of approaches. For small cases exhaustive search works fine: the code specified a tiling and a target, and the program quickly found a rectangle or showed that one did not exist. It turns out that for some rectangles one cannot freely specify the tiling.

For larger cases, what worked well was to divide the rectangle into smaller rectangles, requiring each to be magic or almost-magic. Then fill each sub-rectangle using exhaustive search, having discarded the dominoes already used.

We also used simulated annealing, but this turned out to be of little use.

Pictures of every possibility are provided at www.cs.clemson.edu/∼goddard/papers/dominoMagic/. Here is an example of an 7×8 magic rectangle made up of four 2×7 magic rectangles. Indeed, these four rectangles can be arranged to give a 4×14, and a 2×28 magic rectangle.

0	6	0	6	1	5	1	5
0	6	2	4	1	5	2	4
0	6	0	6	2	4	3	3
4	2	5	1	5	1	4	2
5	1	3	3	3	3	4	2
6	0	6	0	5	1	4	2
6	0	5	1	4	2	3	3

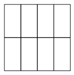
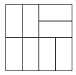
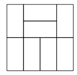
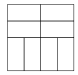
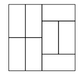
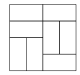
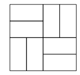
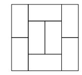
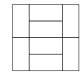
Tiling	1	2	3	4	5	6	7	8	9	
										
Multip.	2	8	4	4	4	8	2	2	2	Total
5	56	26	22	20	30	20	24	48	32	976
6	392	232	270	252	214	202	464	188	440	9384
7	2000	1122	1318	1116	1018	942	1256	1392	1804	43224
8	4488	3028	3466	3044	2916	2839	3832	3256	4344	116480
9	10824	7212	7972	7536	7034	6530	7288	8564	10104	273664
10	14168	11364	12224	12216	11314	10831	13224	11536	13692	425816
11	22896	16606	17574	17928	16744	15233	15824	19512	21936	624032
12	21696	17316	18720	18604	17992	16780	19024	18180	20468	652768
13	22896	16606	17574	17928	16744	15233	15824	19512	21936	624032
14	14168	11364	12224	12216	11314	10831	13224	11536	13692	425816
15	10824	7212	7972	7536	7034	6530	7288	8564	10104	273664
16	4488	3028	3466	3044	2916	2839	3832	3256	4344	116480
17	2000	1122	1318	1116	1018	942	1256	1392	1804	43224
18	392	232	270	252	214	202	464	188	440	9384
19	56	26	22	20	30	20	24	48	32	976
Total	131344	96496	104412	102828	96532	89974	102848	107172	125172	3639920

Table 2: Number of 4×4 domino magic squares by magic sum and dimer tiling

3 Counting Domino Magic Squares

In magic squares the rows, columns and diagonals all sum to the same value. Since there are only 28 dominoes, and the size must be even, and we have already eliminated the 2×2 case, there are only two possibilities: 4×4 and 6×6.

Using exhaustive search we were able to count the number of 4×4 magic squares. We calculated the data by dimer tiling and magic sum. There are 36 dimer tilings of a 4×4 square, but up to rotation and reflection there are only 9 tilings. The data is presented in Table 2. The multiplicity of a tiling is the number of different tilings obtained by rotation and reflection.

Note that every number in the table is even. For 7 of the tilings this is no surprise, since the dimer-tiling has a symmetry such as a reflection or a rotation through 180 degrees, and thus the magic squares are paired off. When the magic sum is 12, then 6-complementation produces another domino magic square, and so again they can easily be paired off. But for the remaining cases it is interesting and curious. And why the hiccup around 12 for some tilings?

As an aside, the number of dimer tilings of a square is well studied; see for example [5]. But we could not find a listing of the count of nonisomorphic tilings.

The 2×2 square has 1, the 4×4 square has 9, and we calculated that the 6×6 square has 930 dimer tilings, but surely this value is known.

4 Larger Squares

It seems likely that one can produce most larger rectangles with larger domino sets. Here is one approach.

The double-R set of dominoes means the set of $(R+2)(R+1)/2$ dominoes with a combination of pips from the range $\{0, 1, \ldots, R\}$. For this set, complements are relative to R, and the self-complementary dominoes are those with pip-sum R. If R is odd, then if one starts with the double-R set and discards the self-complementary dominoes, then one has exactly $(R+1)^2/2$ dominoes left. We can show the following general result:

Lemma 1 *If $R+1$ is a multiple of 8, then the non-self-complementary dominoes in a double-R set can be divided into 2×8 magic rectangles with row-sum $4R$ and column-sum R.*

PROOF. We will place every domino horizontally. Start by pairing each domino with its complement to form a 2×2 square with both columns adding to R.

For each square, let D be half the difference between the row sums. This D ranges from R to 1. Indeed, it follows that the set of D is: R and $R-1$ once, $R-2$ and $R-3$ twice, down to 3 and 2 both $(R-1)/2$ times, and 1 $(R+1)/2$ times (since $R+1$ is a multiple of 4). It follows that for the squares with $D \geq 2$, we can pair them off, each one with a D that is larger or smaller by one: R with $R-1$, and $R-2$ with $R-3$, and so on. Each such pair can be arranged into a 2×4 rectangle with each column adding to R and the rows adding to $2R \pm 1$. Finally, since the number of 2×4 rectangles so formed is even, we can pair them off and place with larger row at the top of one and at the bottom of the other and so form 2×8 magic rectangles.

The number of 2×2 squares with $D = 1$ is a multiple of 4; so these can also be grouped into 2×8 magic rectangles. QED

Now, consider the problem of producing an $n \times n$ domino magic square, n even. Since this needs $n^2/2$ dominoes, one needs at least the double-R set where $R = n-1$. It follows from the above that if n is a multiple of 8, one can construct with the double-$(n-1)$ set an $n \times n$ domino magic rectangle with row-sum and column-sum $n(n-1)/2$. Since there is so much freedom in the construction, surely one can always get the diagonals correct too. For example, here is an 8×8 domino magic square constructed by this approach.

7	7	0	1	0	2	7	4
0	0	7	6	7	5	0	3
6	5	1	1	0	5	7	3
1	2	6	6	7	2	0	4
4	1	4	6	2	2	5	4
3	6	3	1	5	5	2	3
5	1	1	7	2	4	4	4
2	6	6	0	5	3	3	3

References

[1] W.W. Rouse Ball and H.S.M. Coxeter, *Mathematical Recreations and Essays*, Dover, 1987.

[2] Problems P32, P316, P407 and P510 in Henry Dudeney's column in The Weekly Dispatch (see www-cs-faculty.stanford.edu/∼knuth/dudeney-twd.txt)

[3] Ken Duisenberg's Problem of the Week, ken.duisenberg.com/potw

[4] Miodrag Petkovic, *Mathematics and Chess*, Dover Publications, 1997.

[5] N.J.A. Sloane, Sequence A004003 in *The On-Line Encyclopedia of Integer Sequences*, www.research.att.com/∼njas/sequences/A004003